

QIAO FENG

University of Pennsylvania ◊ Philadelphia, PA, USA

(+1) 445 256 2376 ◊ fengqiao@seas.upenn.edu

[Homepage](#) ◊ [Google Scholar](#) ◊ [GitHub](#)

EDUCATION

University of Pennsylvania, Philadelphia, PA, USA

Ph.D. in Computer and Information Science

Aug. 2024 – Present

Advisor: Prof. [Lingjie Liu](#)

Tianjin University, Tianjin, China

M.Eng. in Computer Science and Technology

Sept. 2021 – Jan. 2024

Advisor: Prof. [Kun Li](#)

Tianjin University, Tianjin, China

B.Eng. in Computer Science and Technology

Sept. 2017 – Jul. 2021

GPA: 3.63/4.0

RESEARCH INTERESTS

3D Vision for Robotics. My research develops 3D representations and physics-aware models that enable embodied agents to perceive, simulate, and act in the real world. My work follows a natural progression from *digital humans* to *humanoids* and *robotics*: real-time 3D human reconstruction and generation, physically plausible humanoid motion and control, and general 3D perception and manipulation for robots.

RESEARCH EXPERIENCE

University of Pennsylvania

Ph.D. Student

Aug. 2024 – Present

Advisor: Prof. [Lingjie Liu](#)

Research focuses on 3D vision for embodied AI, spanning digital humans, humanoids, and robotics. Built physics-aware models that connect visual perception with simulation and control, including vision-conditioned humanoid motion reconstruction (*PhysHMR*), text-driven human interaction generation (*Text2Interact*), and video-based reconstruction for mesh-free simulation (*Vid2Sim*). Currently extending this work toward dexterous robot manipulation.

Snap Inc., Snap Research

Research Intern

Jul. 2025 – Oct. 2025

Mentors: [Chuan Guo](#), [Jian Wang](#), [Bing Zhou](#)

Developed motion-aware background video generation methods for avatar-based composition, and built camera-conditioned pipelines for controllable scene synthesis, covering perspective-aware camera control, outpainting, inpainting, and human-scene composition.

Tsinghua University

Visiting Research Assistant

Apr. 2021 – Jul. 2021

Advisor: Prof. [Kun Li](#) & Prof. [Yebin Liu](#)

Worked on animatable avatar reconstruction from monocular video, recovering dynamic human geometry, motion, and appearance from sparse observations.

Tianjin University

Undergraduate & M.Eng. Researcher

Sept. 2019 – Jan. 2024

Advisor: Prof. [Kun Li](#) & Prof. [Yu-Kun Lai](#)

Built the digital-human foundation of my research, working on real-time monocular 3D human reconstruction (*FOF*, *FOF-X*) as well as 3D human generation, editing, and rendering.

(* denotes equal contribution)

1. Qingxuan Wu, Zhiyang Dou, Chuan Guo, Yiming Huang, **Qiao Feng**, Bing Zhou, Jian Wang, Lingjie Liu, “Text2Interact: High-Fidelity and Diverse Text-to-Two-Person Interaction Generation”, in International Conference on Learning Representations (ICLR), 2026. [\[Paper\]](#) [\[Project\]](#) [\[Code\]](#)
2. **Qiao Feng**, Yiming Huang, Yufu Wang, Jiatao Gu, Lingjie Liu, “PhysHMR: Learning Humanoid Control Policies from Vision for Physically Plausible Human Motion Reconstruction”, in ACM SIGGRAPH Asia (SIGGRAPH Asia), 2025. [\[Paper\]](#) [\[Project\]](#) [\[Code\]](#)
3. Chuhao Chen, Zhiyang Dou, Chen Wang, Yiming Huang, Anjun Chen, **Qiao Feng**, Jiatao Gu, Lingjie Liu, “Vid2Sim: Generalizable, Video-based Reconstruction of Appearance, Geometry and Physics for Mesh-free Simulation”, in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025. [\[Project\]](#) [\[Code\]](#)
4. **Qiao Feng**, Yuanwang Yang, Yebin Liu, Yu-Kun Lai, Jingyu Yang, Kun Li, “FOF-X: Towards Real-time Detailed Human Reconstruction from a Single Image”, IEEE Transactions on Multimedia (TMM), 2025. [\[Paper\]](#) [\[Project\]](#) [\[Code\]](#)
5. Yuanwang Yang*, **Qiao Feng***, Yu-Kun Lai, Kun Li, “R²Human: Real-Time 3D Human Appearance Rendering from a Single Image”, in IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2024. [\[Paper\]](#) [\[Project\]](#) [\[Code\]](#)
6. Yuanwang Yang, **Qiao Feng**, Yu-Kun Lai, Kun Li, “Real-time 3D Human Reconstruction and Rendering System from a Single RGB Camera”, in ACM SIGGRAPH Asia Technical Communications, 2024. [\[Project\]](#)
7. Yi Wang*, Jian Ma*, Ruizhi Shao, **Qiao Feng**, Yu-Kun Lai, Yebin Liu, Kun Li, “Human-Coser: Layered 3D Human Generation via Semantic-Aware Diffusion Model”, in IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2024. [\[Project\]](#)
8. Muxin Zhang*, **Qiao Feng***, Zhuo Su, Chao Wen, Zhou Xue, Kun Li, “Joint2Human: High-quality 3D Human Generation via Compact Spherical Embedding of 3D Joints”, in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. [\[Paper\]](#) [\[Project\]](#)
9. Haoyang Ge*, **Qiao Feng***, Hailong Jia, Xiongzheng Li, Xiangjun Yin, You Zhou, Jingyu Yang, Kun Li, “LPSNet: End-to-End Human Pose and Shape Estimation with Lensless Imaging”, in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. [\[Project\]](#)
10. Xiaokun Sun, **Qiao Feng**, Xiongzheng Li, Jinsong Zhang, Yu-Kun Lai, Jingyu Yang, Kun Li, “Learning Semantic-Aware Disentangled Representation for Flexible 3D Human Body Editing”, in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023. [\[Project\]](#) [\[Code\]](#)
11. Xinyi Jing*, **Qiao Feng***, Yu-Kun Lai, Jinsong Zhang, Yuanqiang Yu, Kun Li, “STATE: Learning Structure and Texture Representations for Novel View Synthesis”, Computational Visual Media (CVM), 2023. [\[Project\]](#) [\[Code\]](#)
12. **Qiao Feng**, Yebin Liu, Yu-Kun Lai, Jingyu Yang, Kun Li, “FOF: Learning Fourier Occupancy Field for Monocular Real-time Human Reconstruction”, in Advances in Neural Information Processing Systems (NeurIPS), 2022. [\[Paper\]](#) [\[Project\]](#) [\[Code\]](#)

13. **Qiao Feng**, Yebin Liu, Yu-Kun Lai, Jingyu Yang, Kun Li, “Monocular Real-Time Human Geometry Reconstruction”, in CAAI International Conference on Artificial Intelligence (**CICAI**), 2022. (Demo version of FOF) [\[Paper\]](#)
14. Yi Wang, Jing-Song Cheng, **Qiao Feng**, Wen-Yuan Tao, Yu-Kun Lai, Kun Li, “TSNeRF: Text-driven stylized neural radiance fields via semantic contrastive learning”, in Computers & Graphics, 2023. [\[Paper\]](#)
15. Kun Li, Hao Wen*, **Qiao Feng***, Yuxiang Zhang, Xiongzheng Li, Jing Huang, Cunkuan Yuan, Yu-Kun Lai, Yebin Liu, “Image-Guided Human Reconstruction via Multi-Scale Graph Transformation Networks”, IEEE Transactions on Image Processing (**TIP**), 2021. [\[Project\]](#) [\[Code\]](#)

SCHOLARSHIPS & AWARDS

- Meshy AI Fellowship, Finalist, 2025.
- National Scholarship, Tianjin University, 2022.
- Weichai Scholarship, Tianjin University, 2019.
- Third Prize in the National Student Computer System Capability Challenge (NSCSCC), 2019.
- Competitive Programming
 - Gold medal in the ACM-ICPC Asia Regional Contest, Nanjing Site, 2018.
 - Gold medal in the ACM-ICPC Asia Regional Contest, Qingdao Site, 2018.
 - Silver medal in the ICPC Asia-East Continent Final, Xi'an Site, 2018.
 - First Prize, National Olympiad in Informatics in Provinces (NOIP), 2016.
- First Prize, National High School Math League, 2016.

ACADEMIC SERVICE

- Reviewer: ECCV, ACM Multimedia, ACM SIGGRAPH / SIGGRAPH Asia, IEEE TVCG, IEEE THMS, Pattern Recognition.
- Teaching Assistant, Machine Perception (CIS 5800), University of Pennsylvania, 2025.
- Teaching Assistant, Principles of Programming (Bilingual), Tianjin University, 2019.
- Teaching Assistant, Data Structure, Tianjin University, 2020–2022.
- Teaching Assistant, Algorithms Design and Analysis, Tianjin University, 2020–2022.

SKILLS

- Programming Languages: C/C++, CUDA, Python, Java, Pascal, GLSL
- Tools & Frameworks: PyTorch, Blender, Qt, SolidWorks
- Languages: English (IELTS 7.0), Mandarin Chinese (native)